

Telecom Subscriber Segmentation — Analytics Report

Business Analytics Report

Project: Telecom Subscriber Segmentation

Dataset: telecom_subscribers_usage.csv

Analysis date: 2026-09-25

Prepared by: kabeer ali

Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **telecom_subscribers_usage.csv** (4,000 rows, 17 columns) covering 2016-07-03 to 2026-04-28.

Business question: *A synthetic dataset of 4,020 mobile subscribers across eleven Pakistani cities, with 17 columns covering demographics, plan type, device tier, activation date, monthly data usage, call minutes, SMS, night-time usage, international calls, social media and streaming usage, roaming, recharge amount, and complaints. It has no target column and is intended for customer segmentation. It includes missing values, duplicates, text-formatted numbers, inconsistent labels, impossible ages, and extreme usage outliers.*

Key verified findings

- The original dataset contained 642 missing value(s) and 20 duplicate row(s), which have since been cleaned.
- A Random Forest model was trained to predict **Plan Type** (classification). The model correctly classifies about 87.6% of cases (Accuracy).

Model reliability, in plain terms: The model correctly classifies about 87.6% of cases (Accuracy). F1 score (balancing precision and recall) is 0.873. ROC-AUC is 0.849 (1.0 = perfect separation, 0.5 = no better than chance).

Top recommendations

- The most common observed outcome for Plan Type is 'Prepaid' (606 of 800 test cases).
- Among records, the 'Call Minutes Month: Higher' group shows the highest observed rate (53.1%, n=1334), compared to 'Call Minutes Month: Lower' (8.6%, n=1333).

2. About the Dataset

telecom_subscribers_usage.csv contains 4,000 rows and 17 columns. It spans the period 2016-07-03 to 2026-04-28 (based on Activation Date).

Business question: A synthetic dataset of 4,020 mobile subscribers across eleven Pakistani cities, with 17 columns covering demographics, plan type, device tier, activation date, monthly data usage, call minutes, SMS, night-time usage, international calls, social media and streaming usage, roaming, recharge amount, and complaints. It has no target column and is intended for customer segmentation. It includes missing values, duplicates, text-formatted numbers, inconsistent labels, impossible ages, and extreme usage outliers.

Note: the project description mentions Amount, Target, but no corresponding column was found in this dataset — this report cannot speak to that directly.

Column dictionary

Column	Type	Missing	Notes
Subscriber Id	Identifier	0.0%	
Activation Date	Date/Time	0.0%	
City	Category	0.0%	most common: Karachi
Age	Numeric	0.0%	typical range 7.00–134.00
Gender	Category	0.0%	most common: Male
Plan Type	Category	0.0%	most common: Prepaid
Device Tier	Category	0.0%	most common: Low
Data Usage Gb Month	Numeric	0.75%	typical range 0.10–160.00
Call Minutes Month	Numeric	0.0%	typical range 0.00–71,000.0
Sms Month	Numeric	0.0%	typical range 0.00–165.00
Night Data Share Pct	Numeric	0.0%	typical range 0.00–88.80

International Minutes Month	Numeric	0.0%	typical range 0.00–659.00
Social Media Share Pct	Numeric	0.0%	typical range 0.00–95.00
Video Streaming Hours Month	Numeric	0.0%	typical range 0.00–111.20
Roaming Days Last Year	Numeric	0.0%	typical range 0.00–20.00
Avg Monthly Recharge Pkr	Numeric	0.0%	typical range 110.00–7,320.
Complaints Last 6 Months	Numeric	0.0%	typical range 0.00–6.00

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
4,020	642	20	14

Duplicate Rows

Column	Finding	Affected
(whole dataset)	20 duplicate row(s) detected.	20

Missing Values

Column	Finding	Affected
age	'age' has 160 missing value(s) (4.0%).	160
device_tier	'device_tier' has 121 missing value(s) (3.0%).	121
sms_month	'sms_month' has 201 missing value(s) (5.0%).	201
video_streaming_hours_month	'video_streaming_hours_month' has 160 missing value(s) (4.0%).	160

Impossible Values

Column	Finding	Affected
age	'age' has 1 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	1

Outliers

Column	Finding	Affected
age	'age' has 1 statistical outlier(s) outside [-12.5, 87.5].	1
call_minutes_month	'call_minutes_month' has 774 statistical outlier(s) outside [-151.0, 601.0].	774
sms_month	'sms_month' has 71 statistical outlier(s) outside [-35.0, 109.0].	71
international_minutes_month	'international_minutes_month' has 603 statistical outlier(s) outside [-36.0, 60.0].	603

video_streaming_hours_month	'video_streaming_hours_month' has 7 statistical outlier(s) outside [-59.59, 104.11].	7
roaming_days_last_year	'roaming_days_last_year' has 700 statistical outlier(s) outside [-3.0, 5.0].	700
avg_monthly_recharge_pkr	'avg_monthly_recharge_pkr' has 82 statistical outlier(s) outside [-2485.0, 5475.0].	82
complaints_last_6_months	'complaints_last_6_months' has 58 statistical outlier(s) outside [-1.5, 2.5].	58

Cleaning performed

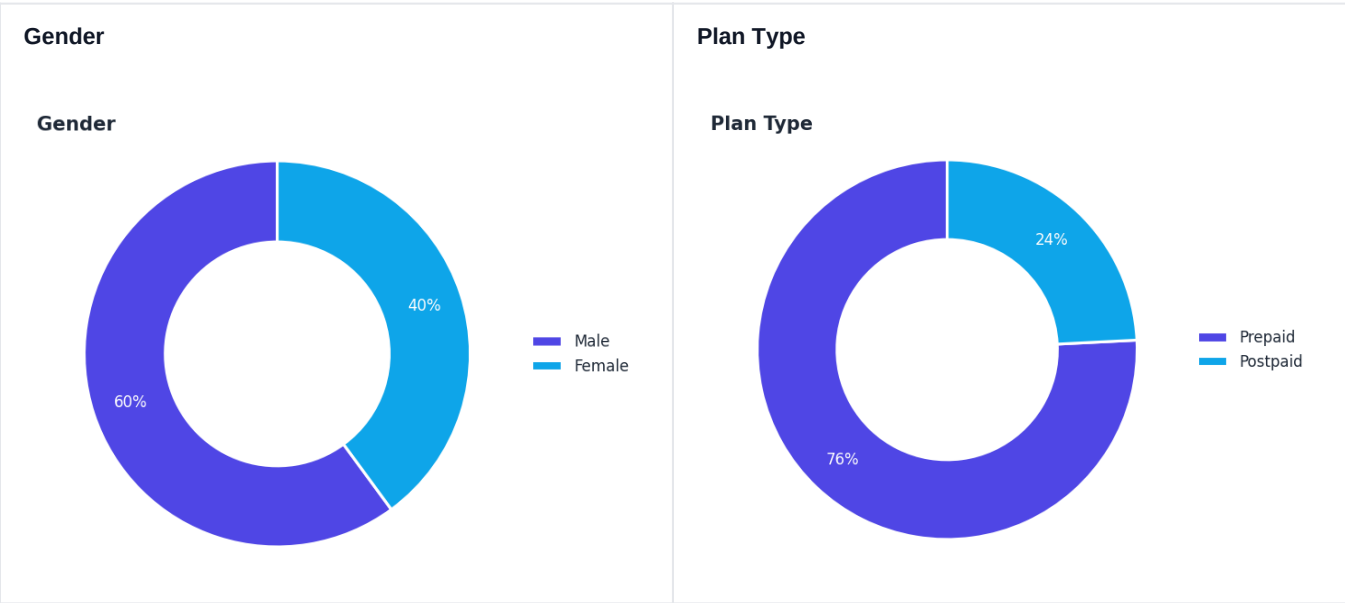
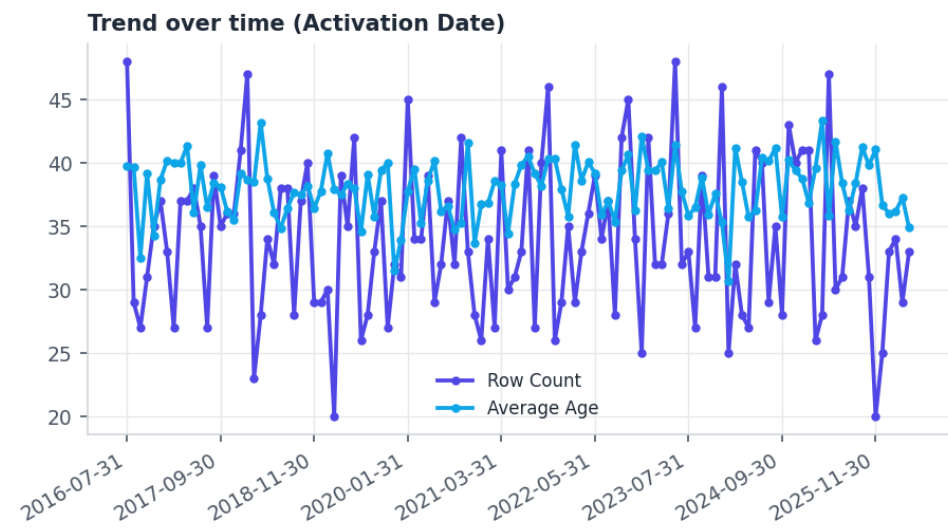
Problem	Before	Action	After	Reason
Duplicate rows	20 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Age	160 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Device Tier	121 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Sms Month	201 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Video Streaming Hours Month	160 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Impossible negative Age	1 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.
Text-formatted numbers in Data Usage Gb Month	stored as text with units (e.g. '4.2 km')	Converted to a clean numeric value	numeric	99% of values matched a number-with-unit pattern.
Inconsistent capitalization in Plan Type	2 variant label(s) (e.g. 'rainy' vs 'Rainy')	Merged into a single canonical label	consistent labels	Values differing only by letter case would otherwise be treated as different categories.

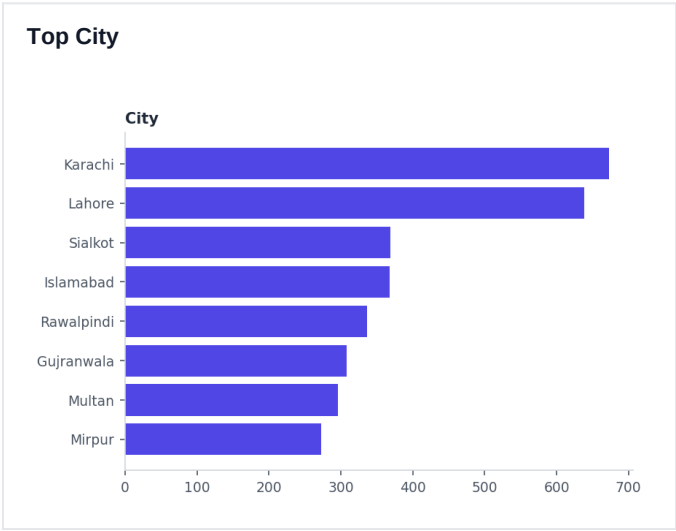
After cleaning: 30 missing cell(s), 0 duplicate row(s), 4,000 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Prepaid Plan Type Rate	Total Age	Total Data Usage Gb Month
4,000	75.8%	152,070	117,825

Trend over time (Activation Date)





Complaints Last 6 Months breakdown

Average: 0.49 (range 0.0–6.0)

Value	Count	Percent
6	1	0.0%
5	2	0.1%
4	7	0.2%
3	48	1.2%
2	303	7.6%
1	1147	28.7%
0	2492	62.3%

5. Modeling Approach

A **classification** model was trained to predict **Plan Type**, using 17 feature(s).

Features used

Activation Date: Year, Activation Date: Month, Activation Date: Dayofweek, City, Age, Gender, Device Tier, Data Usage Gb Month, Call Minutes Month, Sms Month, Night Data Share Pct, International Minutes Month, Social Media Share Pct, Video Streaming Hours Month, Roaming Days Last Year, Avg Monthly Recharge Pkr, Complaints Last 6 Months

Features excluded, and why

Column	Reason excluded
Subscriber Id	Identifier column (e.g. an ID) — carries no predictive signal.
Activation Date	Raw date — replaced with derived year / month / day-of-week features instead.

Re-included at the user's request: Complaints Last 6 Months

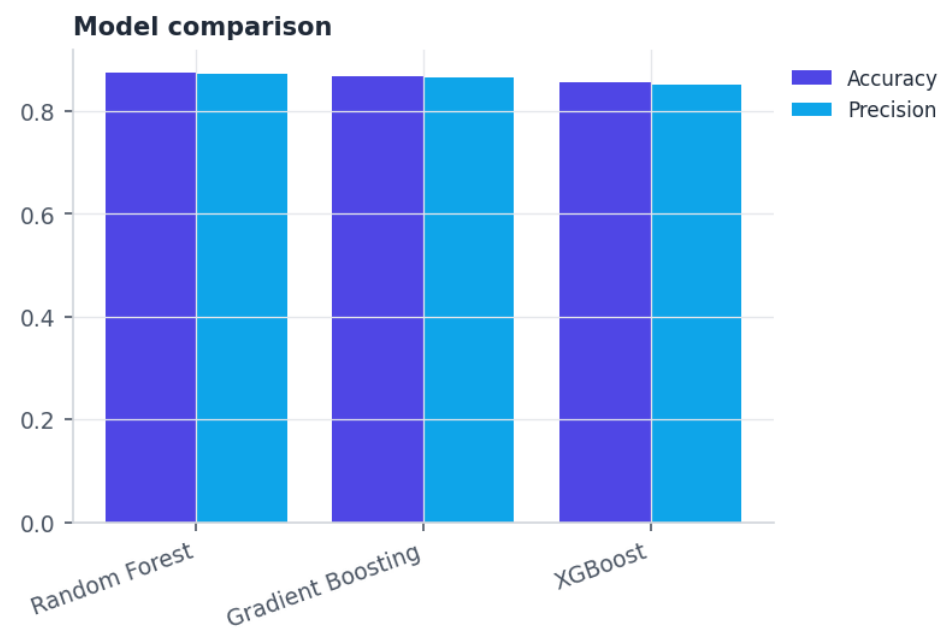
Technical details

Algorithm: Random Forest · Preprocessing: 14 numeric + 3 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.8762	0.8725	0.8762	0.8732	0.8493
Gradient Boosting	0.8688	0.8648	0.8688	0.8658	0.8657
XGBoost	0.8562	0.8511	0.8562	0.8524	0.8398

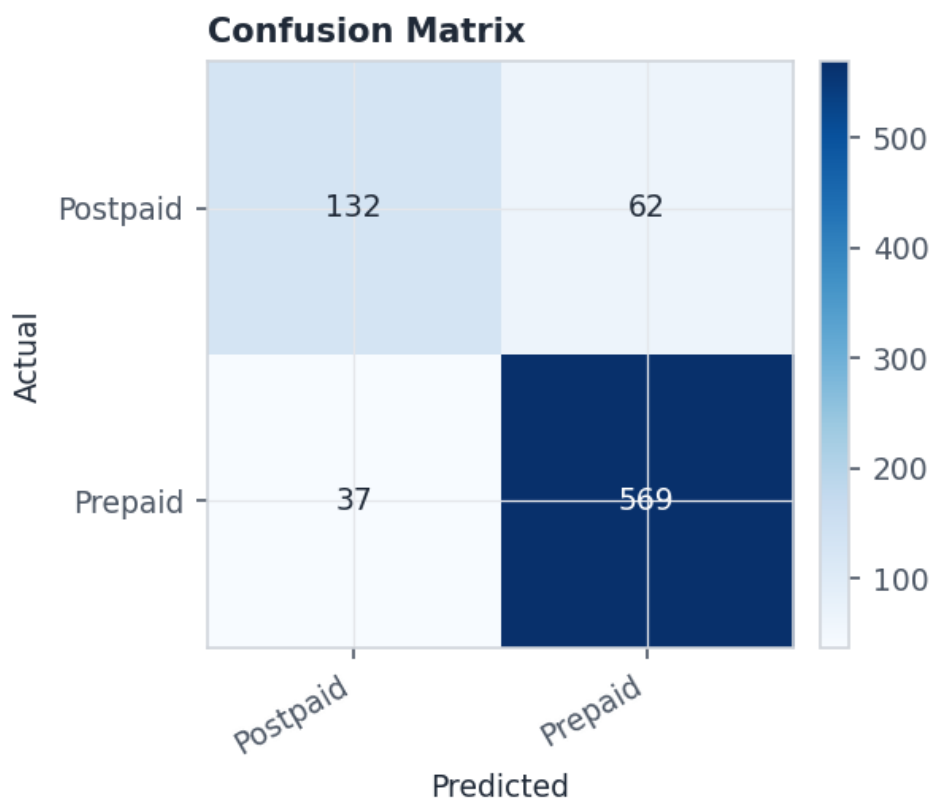


In plain language

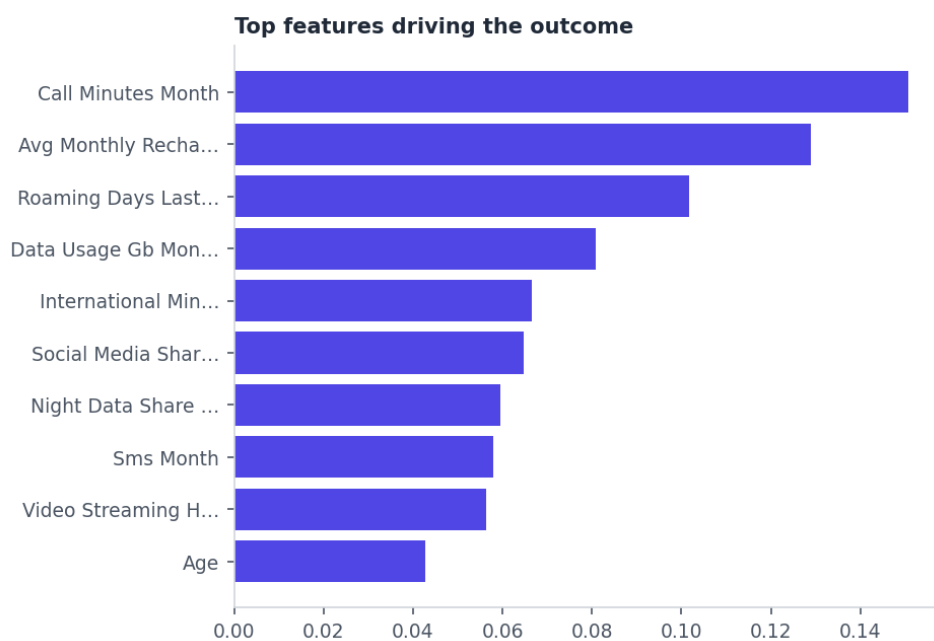
- The model correctly classifies about 87.6% of cases (Accuracy).
- F1 score (balancing precision and recall) is 0.873.
- ROC-AUC is 0.849 (1.0 = perfect separation, 0.5 = no better than chance).

Compared against a trivial baseline (F1 Score = 0.653): this model clearly outperforms it.

Confusion Matrix



7. What Drives the Outcome



Feature	Relative importance
Call Minutes Month	15.1%
Avg Monthly Recharge Pkr	12.9%
Roaming Days Last Year	10.2%
Data Usage Gb Month	8.1%
International Minutes Month	6.7%
Social Media Share Pct	6.5%
Night Data Share Pct	5.9%
Sms Month	5.8%
Video Streaming Hours Month	5.6%
Age	4.3%

8. Future Outlook

- The most common observed outcome for Plan Type is 'Prepaid' (606 of 800 test cases).
- Among records, the 'Call Minutes Month: Higher' group shows the highest observed rate (53.1%, n=1334), compared to 'Call Minutes Month: Lower' (8.6%, n=1333).

Group	Observed rate	Group size
Call Minutes Month: Lower	8.6%	1333
Call Minutes Month: Middle	10.9%	1333
Call Minutes Month: Higher	53.1%	1334

9. AI Business Insights

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- Cross-validation scores varied by ± 0.004 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Accuracy	0.8762
Precision	0.8725
Recall	0.8762
F1 Score	0.8732
ROC-AUC	0.8493
Cv Mean	0.8533
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Class Weight Applied	balanced
Random seed	42
Model version	4

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	4,000
Columns	17